1. Overview

This data wrangling process aims to prepare an analysis-ready encounter-level dataset for answering the research questions. The final dataset will contain a binary outcome variable for postoperative hypoxemia, patient demographics (age, sex, BMI), and the latest available preoperative results for ten specific laboratory tests, namely, leukocytes, pH, hematocrit, CRP, lactate, carbon dioxide, sodium, glucose, hemoglobin, and potassium.

2. Source Data Tables and Relevant Features

Relevant information on patient demographics, lab test results and postoperative hypoxemia status are obtained from three primary data tables from the MOVER dataset [1]. The relevant variables from these tables are outlined in the Data Mapping section. All three data tables are at the encounter level, and share the same unique identifiers for each encounter (namely, the log id (LOG_ID)).

2.1 Patient Postoperative Complications Table

The postoperative complications table contains information on 64,214 encounters. The primary variable of interest within this table is SMRTDTA_ELEM_VALUE, which contains the specific names of postoperative complications.

2.2 Patient Information Table

The patient information table contains information on 64,354 encounters. The descriptions of variables of interest are listed below:

1. BIRTH_DATE: the patient's age in years.
2. SEX: the patient's genotypic sex.
3. HEIGHT: the patient's height in feet and inches (ex. 5' 6).
4. WEIGHT: the patient's weight in ounces.
5. AN_START_DATETIME: the start date and time of anesthesia.

We will rename the BIRTH_DATE variable to AGE for clarity.

2.3 Patient Labs Table

The patient labs table contains information for 29,079,344 lab test results across 56,125 unique encounters. The variables of interest are

1. Lab Code: the unique identifier for the laboratory test.
2. Lab Name: descriptive common name for the laboratory test.
3. Observation Value: observed value (results) for the lab.

4. Measurement Units: units associated with the observed value.
5. Collection Datetime: date and time at which the lab was taken.

3. Data Integration and Restructuring

After selecting the relevant features, we will create three separate dataframes from each of the source tables:

1. Postoperative hypoxemia table: containing LOG_ID, MRN, and postoperative complications
2. Demographics table: containing LOG_ID, MRN, age, sex, and anesthesia start time
3. Lab test results table: containing LOG_ID, MRN, lab code, lab name, value and unit of the results, and collection time

where LOG_ID uniquely identifies each encounter, and MRN (medical record number) uniquely identifies each patient. We identified four LOG_ID values associated with multiple MRNs, indicating single encounters linked to multiple patients. To ensure data integrity, we will exclude encounters corresponding to these four LOG_IDs, so that each retained encounter corresponds to exactly one patient. As the analysis will focus on encounter-level prediction of postoperative hypoxemia, we will then merge the three dataframes by LOG_ID and MRN.

4. Data Cleaning, Transformation and Feature Engineering

4.1 Outcome Variable: Binary Indicator of Postoperative Hypoxemia

We will use the SMRTDTA_ELEM_VALUE (name of postoperative complications) variable to create the binary indicator for postoperative hypoxemia. Entries containing the keyword “hypoxemia” are classified as positive cases, while all other entries are considered negative.

The SMRTDTA_ELEM_VALUE variable has 6,741 out of 203,945 entries being non-null. We assume that these missing entries correspond to cases where no specific postoperative complications were observed. Thus, missing entries are also treated as negative cases.

To ensure the correct identification of positive cases, all text entries in this field will first be converted to lowercase. We will then create a binary variable that assigns a value of 1 to rows where the lowercase SMRTDTA_ELEM_VALUE contains the term “hypoxemia” and a value of 0 otherwise.

4.2 Demographic Features

We will calculate the patient’s BMI as weight in kilograms divided by height in meters squared [2]. To facilitate this calculation, we will first treat the missing values in height and weight then convert height from feet and inches into meters and weight from ounces into kilograms using the following conversion formulas:

1. Height in meters = ((feet * 12) + inches) * 0.0254
2. Weight in kilograms = weight in ounces* 0.0283

Missing values for height and weight are addressed using a two-step procedure. Because a single patient can have multiple encounters, we first attempt to fill missing heights and weights using records from the patient's nearest available encounter. If no available data exists, we plan to use regression imputation to predict the missing values based on the patient's age, sex, and whichever height or weight measurement is available.

We will validate the imputation results by comparing the distribution of height and weight before and after imputation to ensure their forms and shapes remain consistent.

Before attempting imputation, we will inspect the missingness patterns using visualizations such as heatmaps. If the data appears to be MCAR or MNAR where the proposed imputation methods would be biased or inappropriate, we will instead retain only the encounters with no missing values for height and weight.

The SEX variable is binary with values "Female" and "Male", except for one encounter where the value is "Unknown." Since this patient has only a single recorded encounter, the missing value cannot be inferred based on the available data. We will thus drop this encounter from further analysis.

There are no missing values in the age variable.

4.3 Timestamps for Anesthesia and Lab Tests

The anesthesia start time and lab test results collection time are both stored as object variables, which will need to be converted into datetime format to allow for accurate filtering and comparison during subsequent wrangling. For the same purpose, we will only include encounters and lab test instances with non-missing anesthesia start time and collection time, respectively.

4.4 Lab Test Results

The Observation Value variable contains an outlier value of 9999999.0, which we will treat as missing. For Measurement Units, missing values are recorded as "Unknown." We will filter the merged data to exclude all entries with an Observation Value of 9999999.0, regardless of the corresponding unit's availability. We will retain results with missing Measurement Units because certain laboratory tests, such as pH, are unitless in nature.

We will follow the procedure outlined below to select relevant laboratory tests.

Since each test can have multiple Lab Name values, we will choose the most common Lab Name value for each test. This will be done by filtering the merged data for entries with a Lab Name value containing the name of the desired test, ignoring case. We will then calculate the total number of entries with each distinct Lab Name value for that test and choose the Lab Name value with the largest total number of entries.

We will then choose the corresponding Lab Code value for each test by including only entries with a Lab Name value equal to the selected most common Lab Name value for that test. Then, we will calculate the total number of entries with each distinct Lab Code value corresponding to that Lab Name. We will choose the Lab Code value with the largest total number of entries. This will give us a Lab Code-Lab Name pair of values for each test (10 pairs in total). We will then filter the merged data to only include entries containing these 10 Lab Code values and these 10 Lab Name values.

For each encounter, we will restrict the merged data to only include laboratory tests with a Collection Datetime prior to the anesthesia start time. Finally, we will retain only the latest available instance for each laboratory test result prior to anesthesia.

5. Citations

(To be finalized in the final version)

1. MOVER data documentation: https://mover.ics.uci.edu/documentation.html
2. BMI reference: https://my.clevelandclinic.org/health/articles/9464-body-mass-index-bmi